

Varun Srivastava

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EDUCATION

University of California, Berkeley

B.A. Computer Science, Mathematics

Graduated Dec 2023

GPA: 3.9/4

EXPERIENCE

Machine Learning Engineer

AdMarketplace

August 2024 – Present

New York, NY

- Leading training and deployment of the embedding (vector) model powering ad retrieval across all major publishers (such as Firefox and Klarna), serving 500M+ product titles and driving >\$100M in annual revenue.
- Train and optimize CTR/CPC time-series models for the firm's algorithmic bidding system, directly impacting millions of dollars in daily ad spend allocation.
- Built an AI shopping assistant app which is used to demo real-time intent-based ad retrieval to prospective clients.

Founding Engineer

Stork Oracle Network

August 2022 – December 2022

San Francisco, CA

- Designed a high-speed and reliable WebSocket API for live crypto data. The pricefeed served several notable paying DeFi clients, such as Vertex and Apex, which traded millions of dollars daily.

Quantitative Analysis Intern

Dexterity Capital

May 2022 – August 2022

Seattle, WA

- Implemented an evolutionary hyperparameter tuning model for the company's proprietary HFT market making algorithm, trading billions in daily volume; improving search time by > 100x.

PROJECTS

NanoGPT Speedrun / NanoChat | github.com/KellerJordan/modded-nanogpt | github.com/karpathy/nanochat

- Top three competitor in a popular LLM training competition (Modded NanoGPT), with several records involving novel pretraining algorithmic research and low-level GPU optimizations.
- Notable contributions to NanoChat (> 40,000 Github Stars), an affordable end-to-end GPT2 reproduction inspired by the NanoGPT speedrun competition.

Spotify MCP | github.com/varunneal/spotify-mcp

- Built an MCP Server that allows LLMs to control Spotify. Winner at the Anthropic MCP Hackathon (Nov 2024).

Semantic Finder | do-me.github.io/SemanticFinder

- Developed one of the first in-browser vector search engines and a corresponding Chrome extension (> 300 Stars).

RESEARCH & PUBLICATIONS

Machine Learning & Vision Researcher | *Berkeley Visual Computing Lab*

December 2023 – May 2024

- Lee, J., **Srivastava, V.**, Jennings, N., Ng, R. "Theory of Human Tetrachromatic Color Experience." *ACM SIGGRAPH*, 2024. Honorable Mention for Technical Papers Awards, spotlighted at the annual conference.
- Researched the emergence of color vision in humans via neurological models, specifically in the case of esoteric conditions such as human tetrachromacy.

Vision & Neuroscience Researcher | *Na-Ji Biophysics Lab*

September 2019 – April 2020

- Implemented computer vision tools in Matlab for researchers to track the pupil size of rats in video data.
- Correlated terabytes of time-series data with pupil measurements to analyze how rodents react to shock stimulus.

TECHNICAL SKILLS

Languages: Python (PyTorch, Triton, Pandas), CUDA, C/C++, JavaScript, Go, Java, SQL, Matlab.

Frameworks & development: AWS, Databricks, Docker, Kubernetes Node, React, Django.